

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Upload electricity consumption dataset Validate uploaded data for missing or invalid values Store dataset for analysis
FR-2	Data Processing	Clean and preprocess the dataset Organize data by state, month, and year Handle missing and duplicate records
FR-3	Data Visualization	Display monthly electricity consumption trends Display yearly electricity consumption trends Show state-wise electricity consumption comparison
FR-4	Dashboard Interaction	Filter data by state Filter data by month and year Update charts dynamically based on selected filters
FR-5	Trend Analysis	Compare electricity consumption across different periods Analyze seasonal consumption patterns Identify highest and lowest electricity-consuming states
FR-6	COVID-19 Impact Analysis	Compare electricity consumption before, during, and after lockdown Highlight changes in electricity usage due to COVID-19

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should have a simple, interactive, and user-friendly interface that allows users to easily navigate, apply filters, and understand electricity consumption trends.
NFR-2	Security	The electricity consumption dataset should be stored securely, and only authorized users should be allowed to modify or update the data and dashboard.

NFR-3	Reliability	The system should provide accurate, consistent, and error-free analysis and visualizations based on the uploaded dataset.
NFR-4	Performance	The dashboard should load quickly, respond to filters without noticeable delay, and efficiently process large electricity consumption datasets.
NFR-5	Availability	The dashboard should be available whenever users need it and support continuous access through Tableau Public or the deployed platform.
NFR-6	Scalability	The system should support future electricity consumption datasets without requiring major modifications to the dashboard.